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**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 3

Building a Dashboard

Submitted By:

Name: Soham Sen

Reg. No.: 2411021240008

Date: 04/09/2026

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

Experiment No.: 3

Building a Dashboard

Problem Statement

A retail company relies on manually prepared reports to monitor business performance. Develop a Power BI dashboard showing sales, revenue, profit and product performance.

Methodology / Steps Followed

Step 1: Data Collection

- Obtained the retail sales dataset in CSV format from Kaggle.
- The CSV file contains retail transaction information such as:
 - Customer ID
 - Customer Name
 - Product ID
 - Product Name
 - Date
 - Quantity
 - Sales
 - Profit
 - Region
 - State
 - Outlet Type
 - Other retail-related fields
- Created a `Customers` table in MySQL containing customer information.
- The MySQL table contains:
 - Customer ID
 - Customer Name
 - Last Name
 - Date of Birth
 - Segment
- Used **Customer ID** as the common field between the CSV and MySQL data.

Customer_ID	Customer_Name	Last_Name	Date_of_Birth	Segment
CUST000001	Curtis	Krause	1985-04-12	Consumer
CUST000002	Jeremy	Foster	1990-08-25	Corporate
CUST000003	Ariel	Mccann	1988-11-03	Consumer
CUST000004	Colleen	Maynard	1992-01-19	Consumer
CUST000005	Gina	Holden	1983-07-30	Corporate

5 rows in set (0.00 sec)

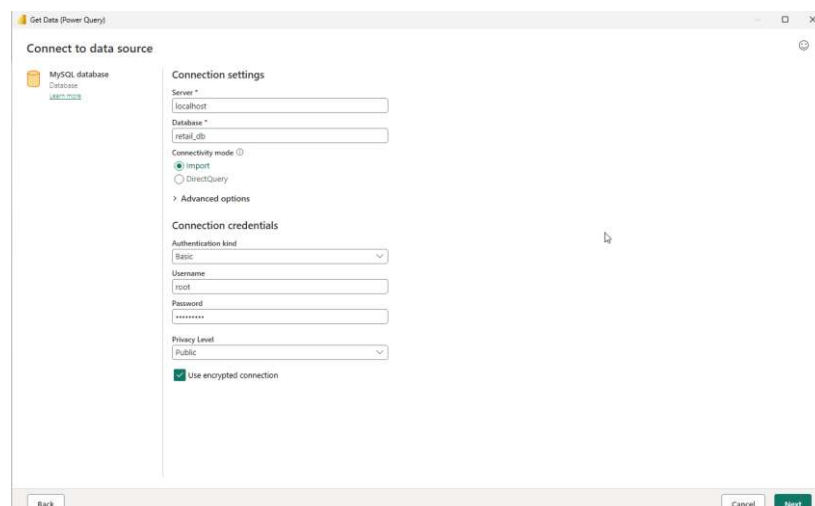
Step 2: Data Import

A. Importing the CSV file

- Open Power BI Desktop.
- Select Home → Get Data.
- Select Text/CSV.
- Select the store_sales_data (2).csv file.
- Preview the dataset.
- Select Transform Data to open Power Query.

B. Importing the MySQL data

- Select Home → Get Data → More.
- Search for MySQL database.
- Select the MySQL connector.
- Enter the MySQL server details.
- Enter the database name:
- retail_db
- Select Import as the storage mode.
- Enter the MySQL username and password.
- Select the Customers table.
- Select Transform Data.



Step 3: Data Cleaning and Transformation

The imported data is checked and prepared using Power Query.

CSV dataset

1. Check the column names.
2. Check for duplicate records.
3. Check for missing values.
4. Verify the data types.
5. Change incorrect data types where necessary.

Column	Data Type
Customer ID	Text
Customer Name	Text
Product ID	Text
Product Name	Text
Date	Date
Quantity	Whole Number
Sales	Decimal Number
Profit	Decimal Number
Region	Text
State	Text
Outlet Type	Text

MySQL Customers table

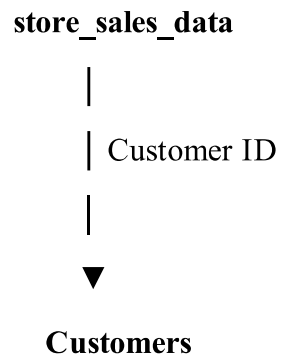
Column	Data Type
Customer_ID	Text
Customer_Name	Text
Last_Name	Text
Date_of_Birth	Date
Segment	Text

6. Rename columns where necessary so that they are easy to understand.
7. Verify that Customer ID and Customer_ID are both stored as Text.
8. Select Close & Apply after completing the transformations.

Step 4: Data Modeling

After loading both datasets, open Model View in Power BI.

The two tables should be:



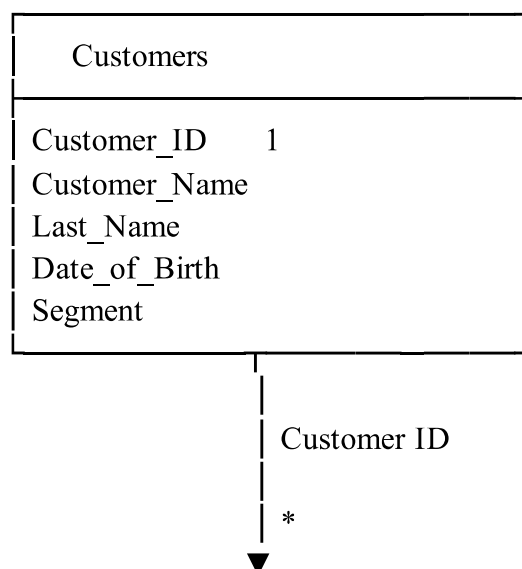
Create the relationship:

Customers[Customer_ID] → store_sales_data[Customer ID]

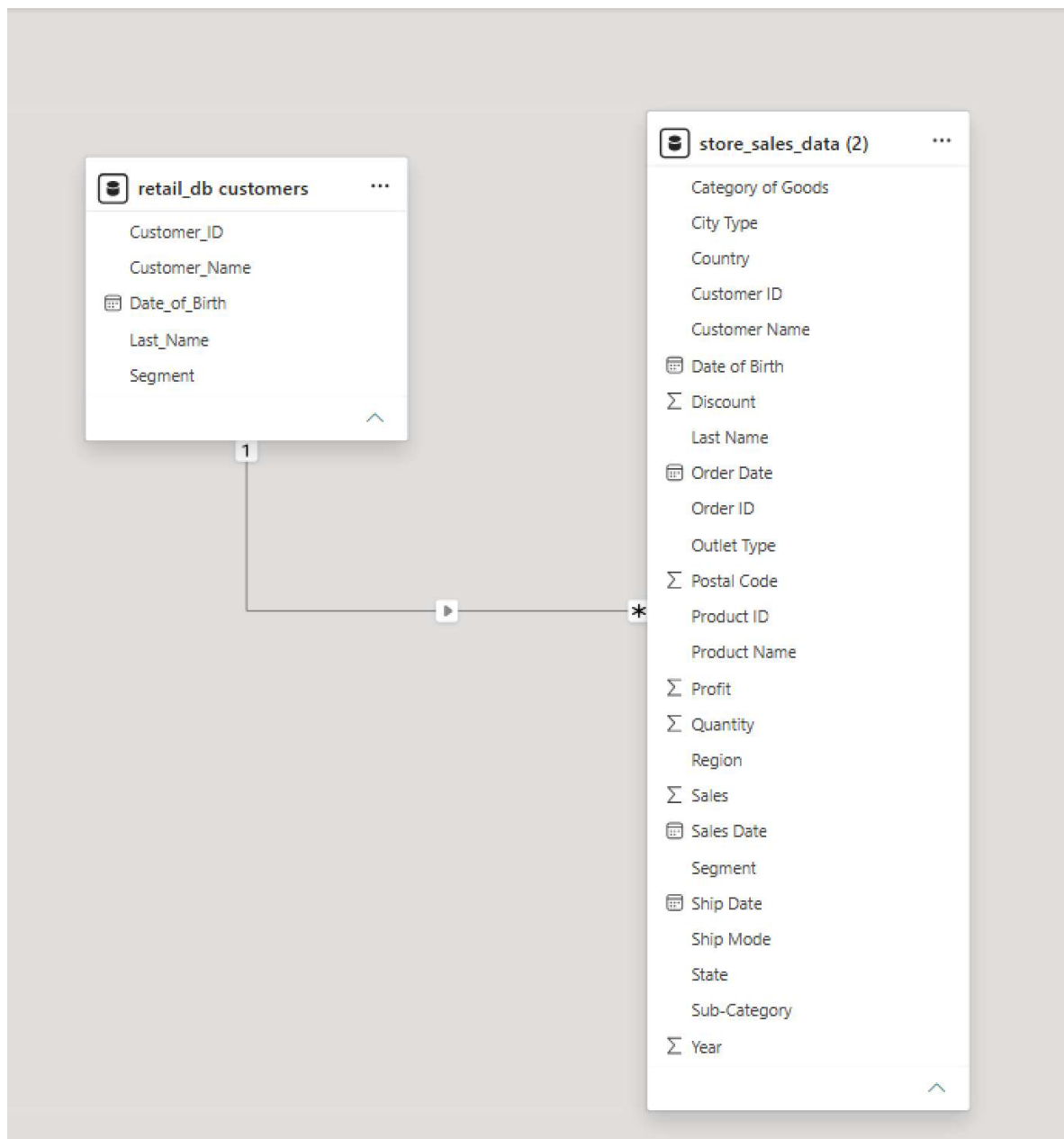
Set:

- Cardinality: One-to-many (1:*)
- Cross-filter direction: Single
- Active: Yes

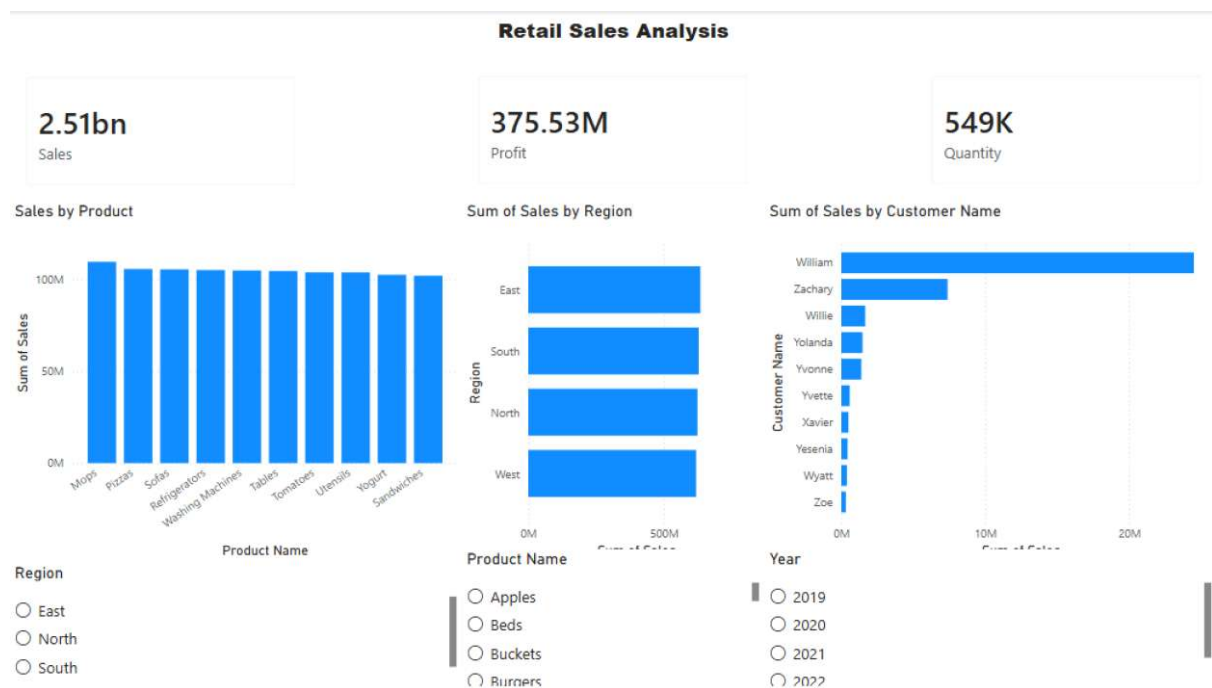
The relationship should look approximately like:



Store Sales Data
Customer ID
Product ID
Product Name
Date
Quantity
Sales
Profit
Region
State
Outlet Type



Output:



The final Power BI dashboard provides:

- Total Sales Performance – Total sales of 2.51 billion.
- Total Profit Analysis – Total profit of 375.53 million.
- Quantity Sold Analysis – Total quantity of 549K units.
- Product-wise Sales Analysis – Comparison of sales across different products such as Mobiles, Pizzas, Sofas, Refrigerators, Washing Machines, Tables, Tomatoes, Utensils, Yogurt, and Sandwiches.
- Regional Sales Performance – Comparison of total sales across the East, South, North, and West regions.
- Customer-wise Sales Analysis – Comparison of sales generated by individual customers.
- Interactive Data Visualization – The charts provide a visual overview of product, regional, and customer sales performance.

The dashboard provides a centralized view of the company's sales, profit, quantity, product performance, regional performance, and customer performance, helping users compare different aspects of retail business performance.

Conclusion

This project successfully integrates retail sales data from a CSV file and customer information from a MySQL database into Power BI. The data was cleaned and transformed using Power Query, and a relationship was established between the two datasets using Customer ID.

The resulting Power BI dashboard provides an interactive view of sales, profit, quantity, customers, products, locations, and time-based trends. This enables business users to analyze retail performance and support data-driven decision-making.

Learning Outcomes

Through this project, I learned:

- How to import CSV files into Power BI.
- How to connect Power BI Desktop to a MySQL database.
- How to use Power Query for data cleaning and transformation.
- How to check and modify data types.
- How to create relationships between tables from different data sources.
- How to use Customer ID as a common relationship key.
- How to create KPI cards and business charts.
- How to analyze sales by product, customer, location, and time.
- How to create interactive slicers and filters.
- How to design an interactive Power BI business dashboard.
- How to generate meaningful business insights from retail data.